DAY 4 · AGENTIC FEST · THE FERGUSON RFQ CHALLENGE

RFQ Agent
from sourcing email to RFQ

One click in Gmail turns a sourcing email into a complete, sourced RFQ in about a minute, flags what is wrong or missing, and lets the team fix it and send it.

👥 Group 3 | 🤖 RFQ Quality Agent | ☁️ Live on Azure

Akepati Thulasi · Promod Belgavi · Raghav Rana · Sanchit Vijay

01 · THE PROBLEM

2 to 4 days of copy and paste for every RFQ

2 to 4 days
to build one RFQ by hand today

~40 a month
RFQs, across 6 countries

23 columns
to fill for every single product

📧 How it works today

A sourcing manager gets an email with product IDs, looks up every product across several sources, and copies each detail into the RFQ Excel before sending it to factories for a price.

⚠️ Where it goes wrong

Copy-paste slips across 23 columns, mixed units (1000 vs 1000 ft vs 333.33 yds), blanks filled by guesswork, competitor part numbers that quietly disappear, and no record of where a value came from.

02 · WHY IT MATTERS

What this means for Ferguson

⚡ Speed

2 to 4 days become about a minute per RFQ. Each worker runs 4 RFQs at once, so nothing waits in line, and factories quote sooner.

🎯 Accuracy

Every value carries its source and a clickable link. Nothing is guessed: unknowns stay blank and are listed with the reason.

📈 Scale

People only review the orange and grey cells instead of typing 23 columns, so the same team handles many more RFQs a month.

💡 **Our one line pitch:** open the RFQ email in Gmail, click once, check the flagged cells, send the RFQ to the factory.

03 · OUR SOLUTION

From email in to RFQ out

Email in
Gmail extension (one click), paste, .eml upload or the inbox bot

→

Find IDs
Rules, not AI: body + Excel/CSV attachments, #IDs, competitor codes, duplicates

→

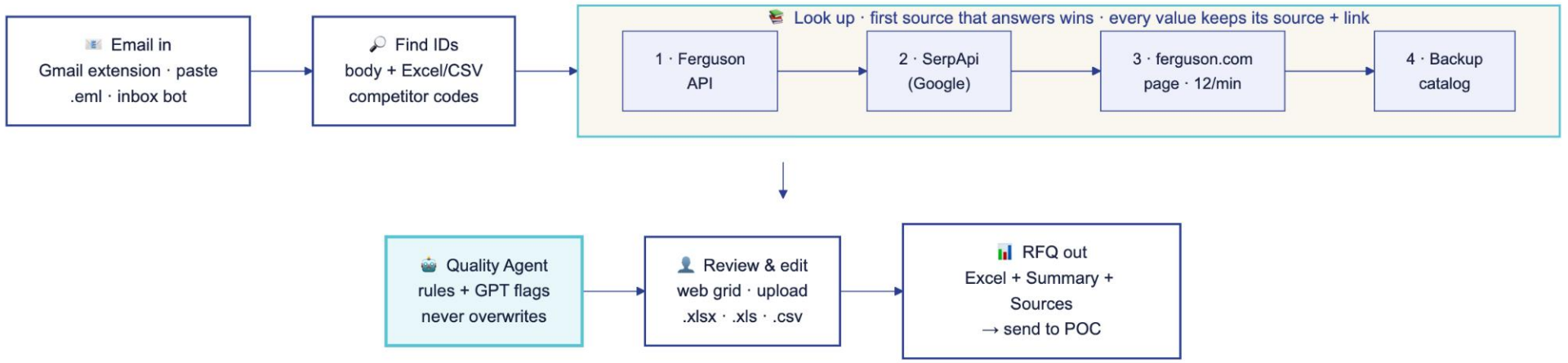
Look up
Ferguson API → SerpApi → ferguson.com page (12/min) → backup catalog

→

Our agentic arm
Units, missing cells, contradictions (rules + GPT); flags, never overwrites

→

RFQ out
Excel with pictures, Sources + links, Summary; review, then send

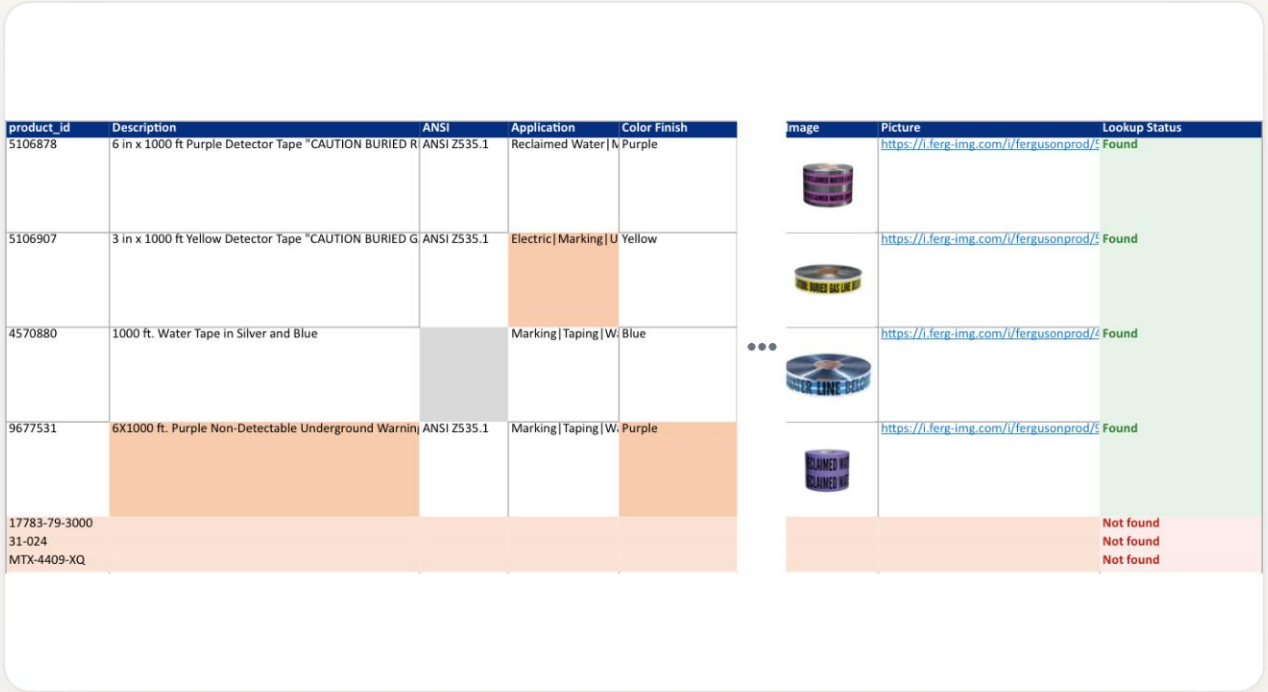


04 · RESULTS

What our flow produced

- Emails 1 to 4: 24 of 28 products filled; the 4 left are competitor part numbers
- Email 1 matches the reference RFQ cell by cell (6 of 6 products)
- "Not found" rows stay on the sheet, blank, with every source tried and the reason
- About 10 to 90 s per RFQ live (rate-limited), ~2 s offline · 121 automated tests

your-azure-app.azurewebsites.net/rfq



product_id	Description	ANSI	Application	Color Finish	Image	Picture	Lookup Status
5106878	6 in x 1000 ft Purple Detector Tape "CAUTION BURIED R	ANSI Z535.1	Reclaimed Water | M	Purple		https://i.ferg-img.com/i/fergusonprod/S	Found
5106907	3 in x 1000 ft Yellow Detector Tape "CAUTION BURIED G	ANSI Z535.1	Electric | Marking | U	Yellow		https://i.ferg-img.com/i/fergusonprod/S	Found
4570880	1000 ft. Water Tape in Silver and Blue		Marking | Taping | W	Blue		https://i.ferg-img.com/i/fergusonprod/4	Found
9677531	6X1000 ft. Purple Non-Detectable Underground Warnin	ANSI Z535.1	Marking | Taping | W	Purple		https://i.ferg-img.com/i/fergusonprod/S	Found
17783-79-3000							Not found
31-024							Not found
MTX-4409-XQ							Not found

05 · OUR AGENTIC ARM

Where the agent made it better

WITHOUT THE AGENT

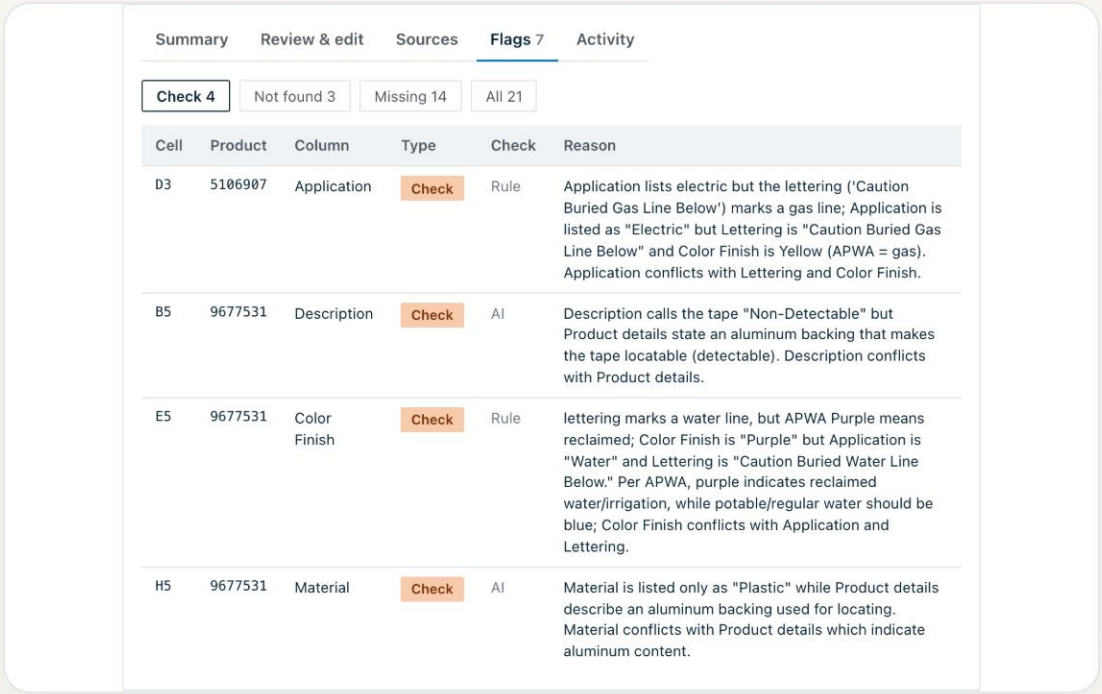
Product 9677531: the catalog values are copied as they are. "Non-Detectable" in the name, aluminium backing in the details, Purple tape for a water line: nobody notices.

WITH THE AGENT

Three cells turn orange, each with its reason: the APWA colour rule flags Purple for a water line, and GPT flags the "Non-Detectable" name and the Material against the aluminium backing. No value is changed.

⚖️ Agent or plain code?

Plain code for what can be checked: units, ranges, colour codes, missing cells. It is fast and testable. GPT only for cross-field contradictions, with strict JSON. It may never write a value, and if it fails the code checks still run.



Summary | Review & edit | Sources | Flags 7 | Activity

Check 4 | Not found 3 | Missing 14 | All 21

Cell	Product	Column	Type	Check	Reason
D3	5106907	Application	Check	Rule	Application lists electric but the lettering ('Caution Buried Gas Line Below') marks a gas line; Application is listed as "Electric" but Lettering is "Caution Buried Gas Line Below" and Color Finish is Yellow (APWA = gas). Application conflicts with Lettering and Color Finish.
B5	9677531	Description	Check	AI	Description calls the tape "Non-Detectable" but Product details state an aluminum backing that makes the tape locatable (detectable). Description conflicts with Product details.
E5	9677531	Color Finish	Check	Rule	lettering marks a water line, but APWA Purple means reclaimed; Color Finish is "Purple" but Application is "Water" and Lettering is "Caution Buried Water Line Below." Per APWA, purple indicates reclaimed water/irrigation, while potable/regular water should be blue; Color Finish conflicts with Application and Lettering.
H5	9677531	Material	Check	AI	Material is listed only as "Plastic" while Product details describe an aluminum backing used for locating. Material conflicts with Product details which indicate aluminum content.

06 · LIMITS AND NEXT STEPS

What we'd build with another week

🚧 Known limits

- The Ferguson API step needs Ferguson API credentials to switch on; until then SerpApi and the rate-limited page read do the work
- Competitor part numbers are flagged, not matched yet; Google does not index every product, and SerpApi can take 20 to 60 s

🚀 Next steps

- Competitor Product Finder as a second arm; switch on the Ferguson product API
- Ferguson SSO instead of a shared key, an Outlook add-in next to Gmail, and the company mail server for sending

📁 **Our code:** github.com/Tanmaya-kh/ferguson-agentic-fest, folder group-3





GROUP 3

Thank you · Questions?

Akepati Thulasi · Promod Belgavi · Raghav Rana · Sanchit Vijay